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In-Database Analytics with SAS: Performance Gains over Traditional Data Movement Approaches

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Abstract—In-Database analytics has been in demand as an effective approach for improving performance of data processing that can handle large and complex datasets. Traditional SAS workflows often require data to be extracted from database system into external analytical environment, can lead to network bottlenecks, increase usage of resources and also reduce scalability. This paper will examine new technologies of SAS In-Database analytics as an alternative approach that pushes computation code directly into the database engine and allows analysis to happen much closer to the data. The discussion of this paper covers the underlying technologies, performance characteristics, model architecture and real-world case study to show on how this approach is able to reduce data movement and improve efficiency. The comparison between traditional SAS workflow and In-database processing will highlight that the In-database approach offer a faster execution, better scalability and lower overhead for large scale analytics. Overall, the paper shows that SAS In-Database analytics is a practical solution for all the organizations that need an efficient and high performance analytics with the massive datasets.

Keywords—In-database analytics, Statistical Analysis Software(SAS), data movement, big data analytics, performance optimization

I. Introduction

In the modern era of big data, most organizations are increasingly required to process and analyze with a massive amount of data efficiently. Traditional data analytics approaches require a system to extract a large datasets from database management systems into analytical environments for processing. While this approach is widely and actively used across the industries, there are several problems that arise including bottleneck network occurrence when moving the data, limited scalability when dealing with large scale datasets and using high cost for data movements.

Therefore, in-database processing like Statistical Analysis Software (SAS) tools has been introduced as a promising that allows data processing directly within database management systems. In-database analytics consists of executing data-analysis and data-mining operations directly within the database engine, so that computations are pushed to the database rather than materializing the full dataset in an external processing environment (Fouché et al., 2018).

This topic is significantly important to enhance performance especially in environments that usually handling large scale and complex datasets. Organizations like Agoda, Walmart, Netflix, Facebook and Starbucks will benefit for using SAS In-Database analytics to support real time decision making and large scale data analysis. This paper will discuss about the architecture of SAS In-Database analysis and compare it with the traditional approach of SAS data processing. The rest of this paper is organized such as follows. Section main body of the paper that discusses about technology, key features, architecture and comparative analysis. The last section is the conclusion that concludes this paper with the following references.

II. Technology

SAS, which stands for Statistical Analysis Software is one of the software analytical tools of In-Database processing that is widely used for data management, advanced analytics, business intelligence and predictive modelling. This software is useful for heavy data analysis when you have vast amounts of information from various sources and it is a smart choice for advanced and complex statistics (Imankhan, 2023).

Traditionally, SAS workflows often rely on using the approach where data will be extracted from databases into SAS environment for processing. However, this approach becomes inefficient due to the network bottleneck and processing delay especially with the data that grows rapidly as the era goes on. That is exactly why in a research paper from D'Silva et al. (2018), technology AIDA or Advanced In-Database Analytics explicitly avoids moving large datasets to the client like in "traditional" SAS processing over SAS/ACCESS. Which is the same motivation behind SAS In-Database that is to push computation to the data to reduce network I/O and latency. SAS In-Database analytics addresses these problems by bring the computation code towards the data.

SAS In-Database processing in a modern day takes a fundamentally different approach. Instead of extracting data from databases and bringing it to SAS environment for processing, this method pushes the SAS code directly into the database engine. The advantage is that SAS allows developers to read, manipulate and analyze large scale datasets using its own programming language which is also called SAS language. After that, the results are sent back to SAS without using a vast amount of resources. Figure 2.1 shows the theory behind in-database processing.

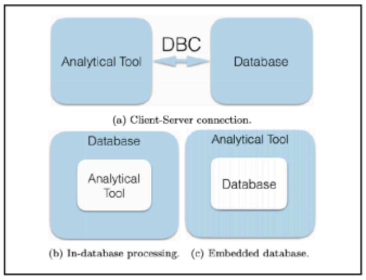

Figure 2.1 In-Database Theory (Amevor, 2020)

III. Key Feature and Performance Characteristics

Table 2.1 below shows the summarization of comparison between traditional SAS workflow and SAS In-Database Processing.

Criteria	Traditional SAS Workflow	SAS In-Database Processing
Data Movement	Large datasets moved to SAS server	Processing occurs inside the database
Performance	Slower for big data	Faster due to reduced data movement
Scalability	Limited by network and memory	Scales with database parallelism
Resource Usage	Heavy load on SAS server	Shared across database nodes
Best Use Case	Small datasets	Large-scale analytics

Table 2.1 Comparison Between Traditional SAS Workflow and SAS In-Database Processing

Traditional RDBMS and classic SAS processing were designed only for small or moderate, structured datasets and also on-premise. As data volume grows, they hit scalability limits and disk-bound bottlenecks when moving large tables into analytic engines (Dritsas & Trigka, 2025). While in-database modern analytics like SAS In-Database analytics push computation inside the database so all the queries and analytics scale better with large volume due to prevention of extracting full table and this process is called in-memory processing.

Studies from Roy et al. (2019) emphasize that in-database geospatial analytics remove the need to extract large spatial datasets and load them into external tools, which reduces network overhead and avoids memory-related crashes when processing the full dataset in-memory. They also note that executing analytic logic within the data warehouse allows faster data processing and minimizes the need to move data back and forth between systems. Abo Khamis et al. (2018) similarly argue that formulating learning tasks as relational queries over sparse tensors inside the DBMS avoids the export loop, which they describe as time-consuming and error-prone, while also enabling a significant performance gains that can outperform existing analytics tools by an order of magnitude in tasks like retail planning.

In conclusion, the traditional SAS workflow usually requires exporting data from database which is time consuming due to its large scale in volume, especially being limited by network and memory. Resulting in lower performance which most organizations want to avoid and preferring using SAS In-Database processing as the main approach.

IV. Architecture Overview

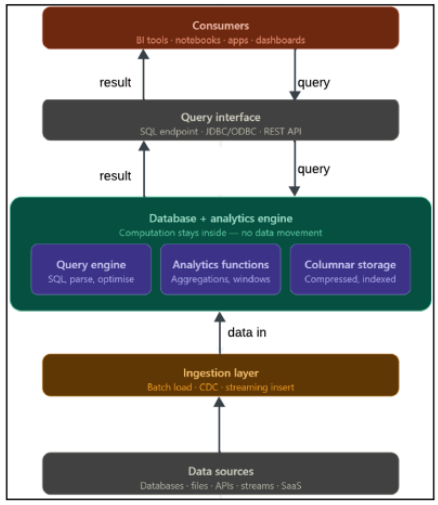

Figure 3.1 In-Database Analytics Architecture

Figure 3.1 shows the architecture that flows from bottom to top through five layers.

First is data sources, the origin of the system including transactional database, files, event stream and SaaS platform. They produce numerous amounts of raw data continuously and transfer everything to the layer above.

Ingestion layer picks up that raw data and loads it into database engine. This can happen as scheduled batch loads, through change data capture (CDC) or via streaming inserts for real-time event data like sensor readings. Data only flows into one direction which data in.

Next, database and analytics engine is the main part of this architecture and the thing that makes this approach distinct from the other. There are three components which are query engine that optimises the SQL sent down from users, analytics function which include aggregations, window functions that execute against stored data and columnar storage that holds the data in a compressed and column-oriented format.

Furthermore, query interface is act like a bridge between the engine and the outside world. It exposes the engine through standard protocols like SQL endpoints, JDBC/ODBC drivers and REST APIs so any tools can connect to this without knowing functionality internally. Lastly, consumers who sit at the top of the architecture can see the result. The result can be seen through a BI tool like Tableau, a Python notebook, a web application, or even a live dashboard.

V. Real-life Case Study

One of the most well known examples of SAS In-Database analytics that are used on a large scale is Walmart which is one of the most largest retail companies in the world. Walmart manages an enormous amount of transactional data daily with millions of point-of-sale transactions that occurring across thousands of stores globally. This company has historically relied on a massive Teradata data warehouse to store and manage this data and the scale of this problem made SAS In-Database processing not just useful but necessary.

VI. Conclusions

This paper concludes with discussion the role of SAS In-Database analytics as a performance oriented alternative towards traditional data movement approaches. The analysis shows that the conventional SAS workflow offer suffers from slower processing, high network overhead and limitation of scalability due to large datasets that need to be transferred out of the database before analysis. Despite that, SAS In-Database processing executes analytics directly inside database engine which can reduce data movement, improve speed and make better use of database parallelism. However, the effectiveness of in-database analytics can still depends on database compatibility, system integration and proper infrastructure support. For the future work can be explore on the research about cloud platforms, advanced machine learning techniques and more optimized hybrid analytics framework.

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13 April 2026	Statistical Analysis Software	what SAS in-database system and what it actually do	Found 2 related research papers. Begin to read
15 April 2026	1. In-Database Analytics 2. Traditional SAS Processing vs. SAS In-Database		Found another 4 research papers. Read the paper and ready to write an introduction and a bit of main body
17 April 2026	Problem with big data analytics		Found another 1 paper about modern database problems. Write a main body (technology and key feature part).
18 April 2026	In-Database Analytics Architecture	What are the element inside in-database analytics architecture	Continue to write a main body content (architecture part) and draw an architecture diagram using Lucidchart.
19 April 2026	In-database SAS real-life case study example	Provide real life example that using in-database SAS.	Write a main body content (real-life case study).
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